

$$1) \quad a - \frac{3}{8} > 0,37$$

$$b - 0,33 < \frac{1}{3}$$

$$c - \sqrt{2} > 1,414$$

$$d - 5 = \sqrt{25}$$

$$e - \frac{3}{7} > 0,428571$$

$$f - \frac{22}{7} > \pi$$

$$1.2) \quad a - 0,(428371)$$

$$b - 7,25$$

$$c - 0,(0693)$$

$$d - 83,12(15)$$

$$1.3) \quad a - \frac{5}{4}$$

$$b - \frac{2374}{1000}$$

$$c - 5 + 0,(3)$$

$$= 5 + \frac{1}{3} = \frac{16}{3}$$

$$d. \text{ Seja } n = 54,134(728)$$

$$10^3 n = 54134, (728)$$

$$10^6 n = 54134728, (728)$$

$$10^6 n - 10^3 n = 54080594$$

$$\Leftrightarrow (10^6 - 10^3)n = 54080594$$

$$\Leftrightarrow n = \frac{54080594}{999000} = \frac{27040297}{499500}$$

$$1.4) a - 0,001 \dots 0,002$$

n° nacional

$$\hookrightarrow 0,00(3)$$

n° internacional

$$\hookrightarrow \frac{\sqrt{2}}{1000}$$

$$b - 0,0099 \dots 0,01$$

n° nacional

$$\hookrightarrow 0,009998$$

n° internacional

$$\hookrightarrow 0,0099(8)$$

1.3) a - falso

$$-3 < 2$$

$$(-3)^2 > 2^2$$

b - Verdadeira

$$x < y$$

$$y^3 - x^3 = \underbrace{(y-x)}_{\oplus} \underbrace{(y^2 + xy + x^2)}_{\oplus}$$

$y^2 > x \vee y$

Como $x < y$ então $0 < y - x$ ou seja $(y-x) > 0$

Por outro lado $(y^2 + xy + x^2) > 0$. Então

$$y^3 - x^3 > 0 \Rightarrow y^3 > x^3$$

c - falso

$$x = 2$$

$$y = 3$$

$$\frac{1}{2} > \frac{1}{3}$$

$$e- \quad x > 7 \Rightarrow |x| > 7$$

$$|x| = \begin{cases} x & \text{se } x \geq 0 \\ -x & \text{se } x < 0 \end{cases}$$

Como $7 > 0$, então 7 é positivo
 logo, se $x > 7$, x é positivo
 por isso $|x| = x$.

Como $x > 7$, então $|x| > 7$

Verdadeiro

$$1.6) a- \quad x, y \in \mathbb{R}$$

$$n \in \mathbb{N}$$

$$\begin{array}{l} \text{para } x = 1 \\ \text{para } y = 2 \\ \text{para } n = 2 \end{array}$$

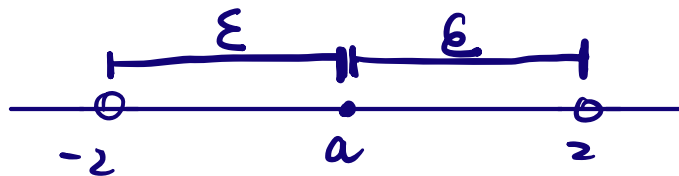
$$(1+2)^2 \neq 1^2 + 2^2$$

$$9 \neq 5$$

Falso

b- Verdadeiro pela multiplicação repetida e
 pelas regras da potência

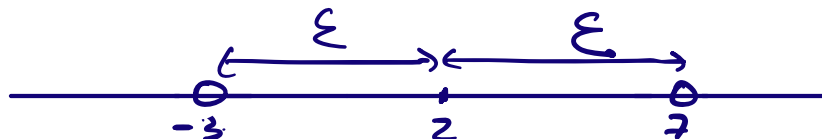
$$1.7) a - |x - a| < \varepsilon$$



$$\text{Centro é } a = \frac{-2+2}{2} = 0$$

$$\text{Raio é } \varepsilon = \frac{2 - (-2)}{2} = \frac{4}{2} = 2$$

d -



$$\text{Centro é } a = \frac{7+(-3)}{2} = \frac{4}{2} = 2$$

$$\text{Raio é } \varepsilon = \frac{7 - (-3)}{2} = 5$$

$$1.8) a - \{x \in \mathbb{R} : |x+4| = 3\}$$

$$|x+4| = \begin{cases} x+4 & \text{se } x \geq 0 \\ -x-4 & \text{se } x < 0 \end{cases}$$

$$\begin{aligned} x+4 &= 3 \\ (\Rightarrow) x &= -1 \end{aligned}$$

$$\begin{aligned} \vee \quad -x-4 &= 3 \\ (\Rightarrow) x &= -7 \end{aligned}$$

$$b - \sqrt{(x+1)^2} = 3$$

$$(\Rightarrow) (x+1)^2 = 3^2$$

$$(\Rightarrow) x^2 + 2x + 1 = 9$$

$$(\Rightarrow) x^2 + 2x - 8 = 0$$

$$(\Rightarrow) x = 2 \vee x = -4$$

$$\sqrt{(x+1)^2} = 3$$

$$(\Rightarrow) |x+1| = 3$$

$$(\Rightarrow) x+1=3 \vee -x-1=3$$

$$(\Rightarrow) x=2 \vee x=-4$$

$$c - |x| = |x+2|$$

$$(\Rightarrow) x = x+2 \vee x = -x-2$$

$$(\Rightarrow) \underbrace{0=2}_{\text{c.f.}} \vee 2x = -2$$

$$(\Rightarrow) \text{---} \quad x = -1$$

$$d - \sqrt{3x+1} = 2x \quad \text{con dominio}$$

$$(\Rightarrow) (\sqrt{3x+1})^2 = 4x^2$$

$$(\Rightarrow) 3x+1 = 4x^2$$

$$(\Rightarrow) -4x^2 + 3x + 1 = 0$$

$$(\Rightarrow) \frac{-3 \pm \sqrt{3^2 - 4(-4)(1)}}{2 \times (-4)}$$

$$(\Rightarrow) x = 1 \vee x = -\frac{1}{4}$$

$$\begin{array}{ll} 3x+1 \geq 0 & \wedge 2x \geq 0 \\ (\Rightarrow) x \geq -\frac{1}{3} & \wedge x \geq 0 \end{array}$$

$$D = \mathbb{R}_0^+$$

$$\text{Come } D = \mathbb{R}_0^+, \quad x=1$$

$$c.s. = 2 \text{ cf}$$

1.9)

$$a - 1 - x \leq 2$$

$$\Leftrightarrow -x \leq 1$$

$$\Leftrightarrow x \geq -1$$

$$C.S = [-1, +\infty[$$

$$c - x^2 > 5$$

$$\Leftrightarrow x > \sqrt{5} \quad \vee \quad x < -\sqrt{5} \quad C.S =]-\infty, -\sqrt{5}[\cup]\sqrt{5}, +\infty[$$

$$e - \left| 5 - \frac{1}{x} \right| < 1 \quad D = (\mathbb{R} \setminus \{0\})$$

$$5 - \frac{1}{x} < 1 \quad \wedge \quad 5 - \frac{1}{x} > -1$$

$$\Leftrightarrow -\frac{1}{x} < -4 \quad \wedge \quad -\frac{1}{x} > -6$$

$$\Leftrightarrow x < \frac{1}{4} \quad \wedge \quad x > \frac{1}{6}$$

$$C.S = \left] \frac{1}{6}, \frac{1}{4} \right[$$

$$f - |3 - x| \geq 2$$

$$\Leftrightarrow 3 - x \geq 2 \quad \vee \quad 3 - x \leq -2$$

$$\Leftrightarrow x \leq 1 \quad \vee \quad x \geq 5$$

$$C.S :]-\infty, 1] \cup [5, +\infty[$$

$$d) |x-1| < |x-2|$$

$$(x-1)^2 < (x-2)^2$$

$$\Rightarrow x^2 - 2x + 1 < x^2 - 4x + 4$$

$$\Rightarrow 2x < 3$$

$$\Leftrightarrow x < \frac{3}{2}$$

$$C.S.]-\infty, \frac{3}{2}[$$

$$e) \frac{x}{x-2} \leq 0$$

$$D = \{x \in \mathbb{R} : x-2 \neq 0\}$$

$$\begin{aligned} x-2 &= 0 \\ x &= 2 \end{aligned}$$

Se x for negativo:

$x-2$ vai ser
sempre negativo
logo $\frac{x}{x-2} > 0$

$$D = \mathbb{R} \setminus \{2\}$$

Por isso, $x > 0$

para x positivo, para que $\frac{x}{x-2} \leq 0$,

$$\begin{aligned} x-2 &\leq 0 \\ \Leftrightarrow x &\leq 2 \end{aligned}$$

$$\frac{x}{x-2} = 0 \Leftrightarrow x = 0 \quad \wedge \quad x \neq 2$$

$$C.S. = [0, 2[$$

$$g - |5n+2| \leq 1$$

$$5n+2 \leq 1 \quad \vee \quad -5n-2 \geq 1$$

$$\Leftrightarrow 5n \leq -1 \quad \vee \quad -5n \geq 3$$

$$\Leftrightarrow n \leq -\frac{1}{5} \quad \vee \quad n \geq -\frac{3}{5}$$

$$C.S = \left[-\frac{3}{5}, -\frac{1}{5}\right]$$

$$h - x^3 \geq 4x$$

$$\Leftrightarrow x^3 - 4x \geq 0$$

$$\Leftrightarrow x(x^2 - 4) \geq 0$$

$$\Leftrightarrow x(x-2)(x+2) \geq 0$$

Para $x(x-2)(x+2)$ ser igual a 0:

$$x=0 \vee x-2=0 \vee x+2=0$$

$$\Leftrightarrow x=0 \vee x=2 \vee x=-2$$

	$-\infty$	-2		0		2	$+\infty$
x	$-$	$-$	$-$	0	$+$	$+$	$+$
$x-2$	$-$	$-$	$-$	$-$	$-$	0	$+$
$x+2$	$-$	0	$+$	$+$	$+$	$+$	$+$
$x(x-2)(x+2)$	$-$	0	$+$	0	$-$	0	$+$

$$C.S = [-2, 0] \cup]2, +\infty[$$

$$i - 6n^2 - 5n \leq -1$$

$$\Leftrightarrow 6n^2 - 5n + 1 \leq 0$$

C.A.

$$6n^2 - 5n + 1 = 0$$

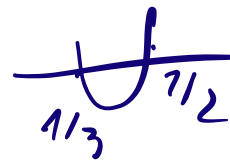
$$\Leftrightarrow n = \frac{5 \pm \sqrt{(-5)^2 - 4 \times 6 \times 1}}{2 \times 6}$$

	$-\infty$	$\frac{1}{3}$		$\frac{1}{2}$	$+\infty$
$6n^2 - 5n + 1$	$+$	0	$-$	0	$+$

$\Rightarrow n = \frac{1}{3} \vee n = \frac{1}{2}$

$$\Leftrightarrow n = \frac{1}{3} \vee n = \frac{1}{2}$$

$$P.S = \left[\frac{1}{3}, \frac{1}{2} \right]$$



$$j - |3n - 2| \leq 1$$

$$\Leftrightarrow 3n - 2 \leq 1 \quad \vee \quad -3n + 2 \geq 1$$

$$\Leftrightarrow 3n \leq 3 \quad \vee \quad -3n \geq -1$$

$$\Leftrightarrow n \leq 1 \quad \vee \quad n \geq \frac{1}{3}$$

$$P.S = \left[\frac{1}{3}, 1 \right]$$

$$k - 2 < |n| < 3$$

$$2 < n < 3 \quad \vee \quad -2 > n > -3$$

$$P.S =]-3, -2[\quad \vee \quad]2, 3[$$

$$M - \frac{1-x}{2x+3} > 0$$

$$D = \{x \in \mathbb{R} : 2x+3 \neq 0\}$$

$$2x+3=0$$

$$\Leftrightarrow 2x = -3$$

$$\Leftrightarrow x = -\frac{3}{2}$$

$$1-x=0$$

$$\Leftrightarrow 1=x$$

$$D = \mathbb{R} \setminus \{-3/2\}$$

	x	$-3/2$		1	x
$1-x$	+	+	+	0	-
$2x+3$	-	0	+	+	+
$\frac{1-x}{2x+3}$	-	n.d.	+	0	-

$$C.S =]-3/2, 1]$$

$$M - |x+2| + |x-2| < 10$$

$$\Leftrightarrow x+2 + x-2 < 10 \quad \vee \quad -x-2 - x+2 > 10$$

$$\Leftrightarrow 2x < 10 \quad \vee \quad -2x > 10$$

$$\Leftrightarrow x < 5 \quad \vee \quad x > -5$$

$$C.S =]-5, 5[$$

$$0 - |x^2 - 1| \leq 1$$

$$(\Leftrightarrow) x^2 - 1 \leq 1 \quad \vee \quad -x^2 + 1 \geq 1$$

$$(\Leftrightarrow) x^2 \leq 2 \quad \vee \quad -x^2 \geq 0$$

$$(\Leftrightarrow) x^2 \leq 2 \quad \vee \quad x^2 \geq 0$$

$$(\Rightarrow) x \leq |\sqrt{2}| \quad \vee \quad x^2 \geq 0$$

$$(\Rightarrow) x \leq \sqrt{2} \quad \wedge \quad x \geq -\sqrt{2} \quad \vee \quad x^2 \geq 0$$

$$C.S = [-\sqrt{2}, \sqrt{2}]$$

$$p - 2x^2 \leq 4$$

$$(\Leftrightarrow) x^2 \leq 2$$

$$(\Leftrightarrow) x \leq |\sqrt{2}|$$

$$(\Leftrightarrow) x \leq \sqrt{2} \quad \wedge \quad x \geq -\sqrt{2}$$

$$C.S = [-\sqrt{2}, \sqrt{2}]$$

$$q - 4 < x^2 < 9$$

$$(\Leftrightarrow) |2| < x < 3$$

$$(\Leftrightarrow) 2 < x < 3 \quad \vee \quad -2 > x > -3$$

$$C.S =]-3, -2[\cup]2, 3[$$

$$1- |x-3| < 2|x|$$

$$\Leftrightarrow |x-3| - 2|x| < 0$$

$$|x-3| = \begin{cases} x-3 & \text{se } x \geq 3 \\ -x+3 & \text{se } x < 3 \end{cases}$$

$$|x| = \begin{cases} x, & \text{se } x \geq 0 \\ -x, & \text{se } x < 0 \end{cases}$$

$$\text{Para } x \geq 3$$

$$|x-3| - 2|x| < 0$$

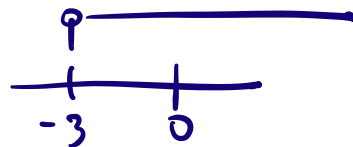
$$\Leftrightarrow x-3-2x < 0$$

$$\Leftrightarrow -x-3 < 0$$

$$\Leftrightarrow -x < 3$$

$$\Leftrightarrow x > -3$$

$$\text{Então } 0 \leq x \leq 3$$



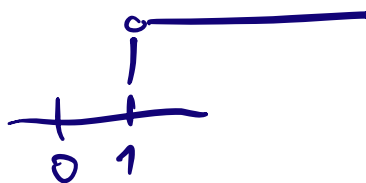
$$|x-3| - 2|x| < 0$$

$$\Leftrightarrow -x+3-2x < 0$$

$$\Leftrightarrow -3x+3 < 0$$

$$\Leftrightarrow -3x < -3$$

$$\Leftrightarrow x > 1$$



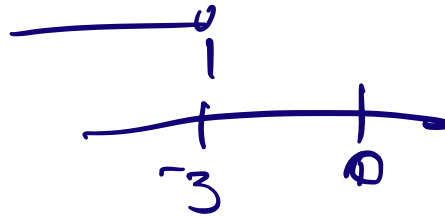
$$\text{find } x < 0$$

$$(x-3) - 2|x| < 0$$

$$\Leftrightarrow -x + 3 + 2x < 0$$

$$\Leftrightarrow x + 3 < 0$$

$$\Leftrightarrow x < -3$$



$$]-\infty, -3[\cap]1, +\infty[\cap]-3, +\infty[$$

$$\text{logo C.S.} =]-\infty, -3[\cap]1, +\infty[$$

$$L - |x+1| > |x-3|$$

$$\Leftrightarrow (x+1)^2 > (x-3)^2$$

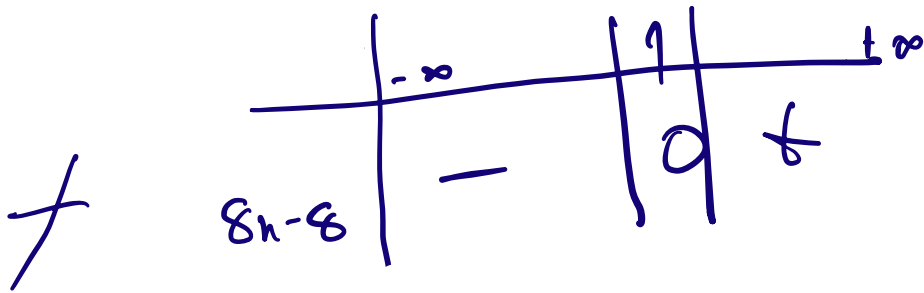
$$\Leftrightarrow \cancel{x^2} + 2x + 1 > \cancel{x^2} - 6x + 9$$

$$\Leftrightarrow 8x - 8 > 0$$

$$g_n - g = 0$$

$$\Leftrightarrow g_n = g$$

$$\Leftrightarrow n = 1$$



$$C.S =]1, +\infty[$$

1.10) a- derivado $\rightarrow \emptyset$

máximos $\rightarrow \bar{n}$ tem

mínimos $\rightarrow \bar{n}$ tem

superior $\rightarrow \bar{n}$ tem

inferior $\rightarrow \bar{n}$ tem

máximo $\rightarrow \bar{n}$ tem

mínimo $\rightarrow \bar{n}$ tem

b - derivado $\rightarrow \mathbb{R}$

2



c - derivado $\rightarrow [0, 2]$

mayorante $\rightarrow [2, +\infty[$

minorante $\rightarrow]-\infty, 0]$

supremo $\rightarrow 2$

infimo $\rightarrow 0$

d - derivado $\rightarrow [-2, 0]$

mayorante $\rightarrow [0, +\infty[$

minorante $\rightarrow]-\infty, -2]$

supremo $\rightarrow 0$

infimo $\rightarrow -2$

mínimo $\rightarrow -2$

h - derivado $\rightarrow \{0\}$

mayorante $\rightarrow [1, +\infty[$

minorante $\rightarrow]-\infty, 0]$

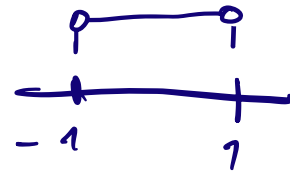
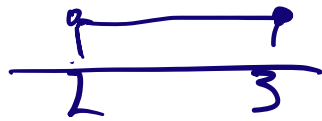
supremo $\rightarrow 1$

infimo $\rightarrow 0$

máximo $\rightarrow 1$

mínimo $\rightarrow n \text{ term}$

2 -



derivado - $[-1, 1] \cup \{2\}$

mayorante - $[3, +\infty[$

menorante - $] -\infty, -1]$

supremo - $\{3\}$

infimo - $\{-1\}$

máximo - 3

mínimo - en term